TUTORIAL 6

Bases, Invertibility, and Elementary Matrices

MOTIVATION AND GUIDING IDEA

Motivation. A basis turns the construction of a vector into the determination of unique coefficients. An invertible matrix guarantees that every input can be reconstructed from its output. Elementary row operations are reversible steps, so row reduction provides both a reconstruction algorithm and an algebraic explanation of why it works.

NOTATION AND CONVENTIONS

Notation guide. For an ordered basis $B = (b_1, \ldots, b_n)$, the notation $[v]_B = (c_1, \ldots, c_n)^T$ means $v = \sum_{j=1}^{n} c_j b_j$. A^{-1} denotes the inverse, A^T the transpose, and $(A \mid I_n)$ an augmented matrix. $R_i \leftrightarrow R_j$ interchanges rows; $R_i \leftarrow \lambda R_i$ requires $\lambda \neq 0$; $R_i \leftarrow R_i + \lambda R_j$ requires $i \neq j$. rank A is the number of pivots in a row-echelon form. Distinguish a matrix unit E_{ij} from an entire elementary matrix E.

DEFINITION 6.1

Hypotheses and tools. Let V have dimension $n < \infty$ over $\mathbb{F}$. Every independent set in V can be extended to a basis; relative to an ordered basis, every $v \in V$ has a unique expression. If $A \in M_n(\mathbb{F})$, $n \geq 1$, is invertible, then $A^{-1} \in M_n(\mathbb{F})$ satisfies $AA^{-1} = A^{-1}A = I_n$. The inverse is unique: if B, C are inverses, then $B = B(AC) = (BA)C = C$.
The following are equivalent: A is invertible; its reduced row-echelon form is I_n; rank $A = n$; $Ax = b$ has a unique solution for every $b \in \mathbb{F}^n$; $Ax = 0$ has only the zero solution; the columns of A form a basis of $\mathbb{F}^n$. For square matrices of the same order, $(AB)^{-1} = B^{-1}A^{-1}$ when both factors are invertible.
If row operations are applied in the order $E_1, E_2, \ldots, E_k$, the result is $E_k \cdots E_2 E_1 A$. If this equals I_n, then $A^{-1} = E_k \cdots E_1$, whereas $A = E_1^{-1} \cdots E_k^{-1}$. The factor order is part of the assertion.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Develop methods for choosing and extending bases, recognize invertible matrices and calculate their inverses, and connect row reduction with products of elementary matrices. Check each solution through its meaning for a system of equations.

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6.1 Bases and Dimension

DEFINITION 6.2

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 6.1

$$\dim(\mathbb{F}^n) = n, \qquad \dim(M_{n\times m}(\mathbb{F})) = nm, \qquad \dim(\mathbb{F}_n[x]) = n+1.$$

Familiar spaces have **standard bases**, denoted $e_1, e_2, \ldots, e_{\dim V}$. For example, in $\mathbb{R}^2$, $e_1 = \begin{pmatrix}1\\0\end{pmatrix}$ and $e_2 = \begin{pmatrix}0\\1\end{pmatrix}$; in $\mathbb{R}_2[x]$, $e_1 = 1, e_2 = x, e_3 = x^2$.

RECALL AND TOOLS

Let V be a vector space over a field $\mathbb{F}$ with finite dimension, $\dim V < \infty$.

1. If B_1, B_2 are bases of V, then $|B_1| = |B_2|$. Thus dimension is well-defined: $\dim V$ is the cardinality of any basis of V.
2. If $W = \{w_1, \ldots, w_m\} \subseteq V$ is linearly independent, then
$$|W| = m \leq \dim V.$$
One cannot choose more independent vectors than the number of vectors in a basis.
3. If $W = \{w_1, \ldots, w_m\} \subseteq V$ spans V, that is, $V = \operatorname{span}(W)$, then
$$|W| = m \geq \dim V.$$
Every spanning set therefore has at least $\dim V$ vectors.
4. Suppose $W = \{w_1, \ldots, w_m\} \subseteq V$ satisfies:
   - $\dim V = |W| = m$;
   - W is either linearly independent or spans V.

   Then W is a basis of V. If W is independent and has exactly $\dim V$ vectors, no further vector can be added while preserving independence, so W already spans. If W spans and has exactly $\dim V$ vectors, there can be no nontrivial zero linear combination of its elements, so they are independent.

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PROPOSITION 6.1

An additional important fact is the following. Let V have dimension n, and let $W = \{w_1, \ldots, w_k\} \subseteq V$ be linearly independent, with $k \leq n$. There is a set

$$U = \{u_{k+1}, \ldots, u_n\} \subseteq V$$

such that

$$B = W \cup U = \{w_1, \ldots, w_k, u_{k+1}, \ldots, u_n\}$$

is a basis of V. Thus **every linearly independent set in a finite-dimensional space can be extended to a basis**. This is a central tool for constructing bases and calculating dimensions.

RECALL AND TOOLS

To establish that a set is a basis of a space of known finite dimension n, it suffices to verify any two of:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

EXERCISE 6.1

Please determine whether

$$v_1 = (1, 0, i)^T, \qquad v_2 = (1+i, 1, 0)^T, \qquad v_3 = (2+i, 1+i, 1+2i)^T$$

form a basis of $\mathbb{C}^3$. If not, please give an explicit dependence relation.

WORKED SOLUTION

For three vectors in $\mathbb{C}^3$, it suffices to test independence. Place them in columns and row-reduce:

$$\begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ i & 0 & 1+2i \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - iR_1} \begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ 0 & 1-i & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (1-i)R_2} \begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ 0 & 0 & 0 \end{pmatrix}.$$

The last step uses $(1-i)(1+i) = 2$. The rank is 2, so the set is not a basis. To identify the relation, verify directly:

$$(2-i)v_1 + (1+i)v_2 = \begin{pmatrix} (2-i) + (1+i)^2 \\ 1+i \\ (2-i)i \end{pmatrix} = \begin{pmatrix} 2+i \\ 1+i \\ 1+2i \end{pmatrix} = v_3.$$

Thus $(2-i)v_1 + (1+i)v_2 - v_3 = 0$ is a nontrivial relation confirming the row reduction.

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EXERCISE 6.2

Please determine whether $B = (x^2 - 1,\ 2x + 3,\ 1 + x + x^2)$ is a basis of $\mathbb{R}_2[x]$, and then whether it is a basis of $\mathbb{R}_3[x]$. If it is a basis in the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Our convention is that $\mathbb{R}_j[x]$ consists of polynomials of degree at most j. Seek α, β, γ such that

$$ax^2 + bx + c = \alpha(x^2 - 1) + \beta(2x + 3) + \gamma(1 + x + x^2).$$

Comparing coefficients gives

$$\alpha + \gamma = a, \qquad 2\beta + \gamma = b, \qquad -\alpha + 3\beta + \gamma = c.$$

The second equation gives $\gamma = b - 2\beta$, and the first gives $\alpha = a - b + 2\beta$. Substitute in the third:

$$-(a - b + 2\beta) + 3\beta + b - 2\beta = c \iff -a + 2b - \beta = c.$$

Consequently,

$$\beta = -a + 2b - c, \quad \gamma = 2a - 3b + 2c, \quad \alpha = -a + 3b - 2c.$$

For every a, b, c there is a unique solution, so B is a basis of $\mathbb{R}_2[x]$, with

$$[p]_B = (-a + 3b - 2c,\ -a + 2b - c,\ 2a - 3b + 2c)^T.$$

However, B does not span $\mathbb{R}_3[x]$: every linear combination of its members has degree at most 2, and therefore cannot equal x^3.

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EXERCISE 6.3

Work over $\mathbb{R}$. The same proof holds over any field $\mathbb{F}$ of characteristic different from 2, after changing the scalar field in all definitions. Let

$$M_4 := M_{4\times 4}(\mathbb{R})$$

be the space of all real 4×4 matrices. Let E_{ij} have 1 in entry (i, j) and zero elsewhere. Define two subspaces of M_4:

$$\mathrm{Sym}_4(\mathbb{R}) := \{A \in M_4 : A^T = A\} \quad \text{(symmetric matrices)},$$

$$\mathrm{ASym}_4(\mathbb{R}) := \{A \in M_4 : A^T = -A\} \quad \text{(skew-symmetric matrices)}.$$

Define

$$D_i := E_{ii} \quad (1 \le i \le 4),$$
$$S_{ij} := E_{ij} + E_{ji} \quad (1 \le i < j \le 4),$$
$$K_{ij} := E_{ij} - E_{ji} \quad (1 \le i < j \le 4).$$

1. Please show that

$$\mathcal{B}_{\mathrm{sym}} := \{D_1, D_2, D_3, D_4,\ S_{12}, S_{13}, S_{14}, S_{23}, S_{24}, S_{34}\}$$

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EXERCISE 6.3 CONTINUED

is a basis of $\mathrm{Sym}_4(\mathbb{R})$.

2. Please show that
$$\mathcal{B}_{\text{skew}} := \{K_{12}, K_{13}, K_{14}, K_{23}, K_{24}, K_{34}\}$$
is a basis of $\mathrm{ASym}_4(\mathbb{R})$.

3. Please show that the union
$$\mathcal{B} := \mathcal{B}_{\text{sym}} \cup \mathcal{B}_{\text{skew}}$$
is a basis of M_4. Thus a basis of the subspace $\mathrm{Sym}_4(\mathbb{R})$ is extended to a basis of all of M_4 by adding vectors from $\mathrm{ASym}_4(\mathbb{R})$.

4. Please compute
$$\dim \mathrm{Sym}_4(\mathbb{R}), \quad \dim \mathrm{ASym}_4(\mathbb{R}), \quad \dim M_4,$$
and please verify
$$\dim \mathrm{Sym}_4(\mathbb{R}) + \dim \mathrm{ASym}_4(\mathbb{R}) = \dim M_4.$$

WORKED SOLUTION

1. $\mathcal{B}_{\text{sym}}$ **is a basis of** $\mathrm{Sym}_4(\mathbb{R})$**.** First, every member of the set is symmetric:
$$D_i^T = D_i, \qquad S_{ij}^T = (E_{ij} + E_{ji})^T = E_{ji} + E_{ij} = S_{ij}.$$
Thus $\mathcal{B}_{\text{sym}} \subseteq \mathrm{Sym}_4(\mathbb{R})$.

Spanning. Let $A = (a_{kl}) \in \mathrm{Sym}_4(\mathbb{R})$. For $1 \le i \le 4$, the diagonal entry a_{ii} is supplied by $a_{ii}D_i$. For $1 \le i < j \le 4$, $a_{ij} = a_{ji}$, and this pair of entries is supplied by $a_{ij}S_{ij}$. Therefore
$$A = \sum_{i=1}^{4} a_{ii} D_i + \sum_{1 \le i < j \le 4} a_{ij} S_{ij}.$$
Every A is a linear combination of members of $\mathcal{B}_{\text{sym}}$, so $\mathrm{Sym}_4(\mathbb{R}) \subseteq \mathrm{span}(\mathcal{B}_{\text{sym}})$.

Independence. Suppose
$$\sum_{i=1}^{4} \alpha_i D_i + \sum_{1 \le i < j \le 4} \beta_{ij} S_{ij} = 0.$$
Only D_i contributes to entry (i,i), giving $\alpha_i = 0$ for every i. For $i < j$, only S_{ij} contributes to entry (i,j), whose value is β_{ij}. Equality with the zero matrix therefore gives $\beta_{ij} = 0$ for every $i < j$. All coefficients vanish, so $\mathcal{B}_{\text{sym}}$ is independent. Since it also spans, it is a basis. Moreover,
$$|\mathcal{B}_{\text{sym}}| = 4 + 6 = 10, \qquad \dim \mathrm{Sym}_4(\mathbb{R}) = 10.$$

2. $\mathcal{B}_{\text{skew}}$ **is a basis of** $\mathrm{ASym}_4(\mathbb{R})$**.** Each member is skew-symmetric:
$$K_{ij}^T = (E_{ij} - E_{ji})^T = E_{ji} - E_{ij} = -K_{ij},$$
so $K_{ij} \in \mathrm{ASym}_4(\mathbb{R})$.

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EXERCISE 6.3 CONTINUED

Spanning. If $A = (a_{kl}) \in \mathrm{ASym}_4(\mathbb{R})$, then

$$a_{ii} = 0 \quad \text{for every } i, \qquad a_{ji} = -a_{ij} \quad \text{for } i < j.$$

It is therefore enough to specify the entries a_{ij} with $1 \le i < j \le 4$; the entries below the diagonal are then determined. We can write

$$A = \sum_{1 \le i < j \le 4} a_{ij} K_{ij},$$

because $K_{ij} = E_{ij} - E_{ji}$ supplies exactly the entries (i, j) and (j, i) with opposite signs. Thus every $A \in \mathrm{ASym}_4(\mathbb{R})$ lies in $\mathrm{span}(\mathcal{B}_{\mathrm{skew}})$.

Independence. Suppose

$$\sum_{1 \le i < j \le 4} \gamma_{ij} K_{ij} = 0.$$

For $i < j$, entry (i, j) is exactly γ_{ij}. Equality with the zero matrix forces $\gamma_{ij} = 0$ for every $i < j$. All coefficients vanish, so the set is independent. Its size is

$$|\mathcal{B}_{\mathrm{skew}}| = 6, \qquad \dim \mathrm{ASym}_4(\mathbb{R}) = 6.$$

3. **The union $\mathcal{B}$ is a basis of M_4 and extends the first basis.** The conceptual facts are:

   - $\mathcal{B}_{\mathrm{sym}}$ is a basis of $\mathrm{Sym}_4(\mathbb{R})$;
   - $\mathcal{B}_{\mathrm{skew}}$ is a basis of $\mathrm{ASym}_4(\mathbb{R})$;
   - the intersection of these subspaces is exactly $\{0\}$.

   Recall the standard decomposition. For each $A \in M_4$, define

   $$S := \frac{1}{2}(A + A^T), \qquad K := \frac{1}{2}(A - A^T).$$

   Then

   $$S^T = \frac{1}{2}(A^T + A) = S, \qquad K^T = \frac{1}{2}(A^T - A) = -K.$$

   Thus $S \in \mathrm{Sym}_4(\mathbb{R})$ and $K \in \mathrm{ASym}_4(\mathbb{R})$, with $A = S + K$. Consequently,

   $$M_4 = \mathrm{Sym}_4(\mathbb{R}) + \mathrm{ASym}_4(\mathbb{R}).$$

   For the intersection, if A is in both summands, then $A^T = A$ and $A^T = -A$, giving

   $$A = -A \implies 2A = 0 \implies A = 0,$$

   since we work over $\mathbb{R}$, where $2 \neq 0$. Hence

   $$\mathrm{Sym}_4(\mathbb{R}) \cap \mathrm{ASym}_4(\mathbb{R}) = \{0\},$$

   and

   $$M_4 = \mathrm{Sym}_4(\mathbb{R}) \oplus \mathrm{ASym}_4(\mathbb{R}).$$

   For the basis conclusion, use the three established facts:

   - $\mathcal{B}_{\mathrm{sym}}$ is a basis of $\mathrm{Sym}_4(\mathbb{R})$;

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EXERCISE 6.3 CONTINUED

- $\mathcal{B}_{\text{skew}}$ is a basis of $\text{ASym}_4(\mathbb{R})$;
- $\text{Sym}_4(\mathbb{R}) \cap \text{ASym}_4(\mathbb{R}) = \{0\}$.

Their union $\mathcal{B} = \mathcal{B}_{\text{sym}} \cup \mathcal{B}_{\text{skew}}$ is therefore a basis of the direct sum, hence a basis of all of M_4. In other words:

- We started with a basis $\mathcal{B}_{\text{sym}}$ of the subspace of symmetric matrices.
- We extended it to a basis of all of M_4 by adding the members of $\mathcal{B}_{\text{skew}}$.
- This concretely illustrates the theorem that every basis of a subspace can be extended to a basis of the whole space.

4. We already found

$$\dim \text{Sym}_4(\mathbb{R}) = 10, \qquad \dim \text{ASym}_4(\mathbb{R}) = 6.$$

Since M_4 contains all 4×4 matrices, $\dim M_4 = 16$. Indeed,

$$\dim \text{Sym}_4(\mathbb{R}) + \dim \text{ASym}_4(\mathbb{R}) = 10 + 6 = 16 = \dim M_4.$$

This is exactly the dimension relation for the direct decomposition

$$M_4 = \text{Sym}_4(\mathbb{R}) \oplus \text{ASym}_4(\mathbb{R}).$$

Interpretation.

- A symmetric matrix is specified by its 4 diagonal entries and its 6 entries above the diagonal, giving 10 free parameters.
- A skew-symmetric matrix has zero diagonal, and the entries below the diagonal are the negatives of those above, giving 6 free parameters.
- Together there are $10+6$ parameters, exactly the number of entries in a 4×4 matrix.
- The combined basis $\mathcal{B}$ corresponds to decomposing each matrix unit E_{ij} into symmetric and skew-symmetric parts.

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6.2 The Dimension Formula

PROPOSITION 6.2

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

The dimension formula:

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PROPOSITION 6.3

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 6.4

1. Let V have dimension n over $\mathbb{F}$, and let $U, W \subseteq V$ be subspaces such that $\dim U = n-1$ and $W \nsubseteq U$. Please prove that:
   (a) $V = U + W$.
   (b) $\dim(U \cap W) = \dim W - 1$.
2. Let $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2+1\}$, and $W = \{P \in V : P'(1) = 0\}$.
   (a) Please compute $\dim(U+W)$ and $\dim(U \cap W)$.
   (b) Is $U + W = V$? If so, is the sum direct, that is, is $U \oplus W = V$?

WORKED SOLUTION

1. (a) By assumption, there is $x \in W \setminus U$. Hence $U \subsetneq U + W \subseteq V$, and the preceding dimension lemma gives

$$n - 1 = \dim U < \dim(U+W) \leq \dim V = n.$$

   Thus $\dim(U+W) = n$, and the equality case in the lemma gives $V = U + W$.

   (b) Apply the dimension formula:

$$\begin{aligned} \dim(U+W) &= \dim U + \dim W - \dim(U \cap W), \\ n &= n - 1 + \dim W - \dim(U \cap W). \end{aligned}$$

   Therefore $\dim(U \cap W) = \dim W - 1$, as required.

2. (a) We have $\dim V = 3$ and $\dim U = 2$, since its displayed spanning vectors are independent. Also $W \nsubseteq U$: for instance, $Q(X) = 1$ belongs to $W \setminus U$. All hypotheses of part 1 hold, giving $V = U + W$ and hence $\dim(U+W) = \dim V = 3$. The second conclusion of part 1 gives $\dim(U \cap W) = \dim W - 1$, so it remains to find $\dim W$. For $P(X) = aX^2 + bX + c \in W$, with $a, b, c \in \mathbb{R}$, the condition is $P'(1) = 2a + b = 0$, that is, $b = -2a$. Thus

$$\begin{aligned} W &= \{aX^2 - 2aX + c : a, c \in \mathbb{R}\} \\ &= \{a(X^2 - 2X) + c \cdot 1 : a, c \in \mathbb{R}\} \\ &= \operatorname{span}\{1, X^2 - 2X\}. \end{aligned}$$

   Consequently $\dim W = 2$, and $\dim(U \cap W) = 1$.

   (b) This is indeed a sum equal to V: $U + W \subseteq V$, and $\dim(U+W) = \dim V$ implies $U + W = V$. The sum is not direct, since

$$(X-1)^2 = (X^2+1) - 2X \in U, \qquad \frac{d}{dX}(X-1)^2\Big|_{X=1} = 0,$$

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EXERCISE 6.4 CONTINUED

so $(X-1)^2 \in W$. In fact $U \cap W = \text{span}\{(X-1)^2\}$, consistent with the computed dimension 1.

MOTIVATION AND GUIDING IDEA

From a basis to unique reconstruction. The columns of a square matrix form a basis if and only if every right-hand-side vector can be expressed uniquely using those columns. This is the meaning of invertibility and explains why an inverse solves a system for every right-hand side.

6.3 Invertible Matrices

DEFINITION 6.3

A matrix $A \in M_n(\mathbb{F})$, with $n \geq 1$, is **invertible** if there is $B \in M_n(\mathbb{F})$ satisfying $AB = BA = I_n$.

The unique inverse B is denoted A^{-1}.

Basic Properties

RECALL AND TOOLS

- If A is invertible, so is A^{-1}.
- If A, B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is invertible, so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.
- The identity is invertible, with $I^{-1} = I$.
- A square zero matrix of positive order is not invertible.

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6.3.1 Exercises on the Algebra of Invertible Matrices

EXERCISE 6.5

Let A, B be square matrices, with A invertible, such that $A = B^2$ and $B = A^2$. Please prove that B is the inverse of A, that is, $B = A^{-1}$.

WORKED SOLUTION

We have

$$\begin{aligned} AB &= AA^2 = A^3, \\ A^3 &= A^3(AA^{-1}) = A^4A^{-1} \\ &= (A^2A^2)A^{-1} = B^2A^{-1} = AA^{-1} = I. \end{aligned}$$

Since $B = A^2$, also $BA = A^2A = A^3 = AB = I$. Therefore $AB = BA = I$, and $B = A^{-1}$.

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EXERCISE 6.6

Let A, B be square matrices such that AB is invertible. Please prove that both A and B are invertible. **This is the converse of the product assertion from the lecture.**

WORKED SOLUTION

Use the fact that for square matrices of the same order, a one-sided inverse is also a two-sided inverse. Indeed, $AX = I$ implies $n = \text{rank}(AX) \leq \text{rank}(A) \leq n$, so A is invertible. From $(AB)(AB)^{-1} = I$, obtain $A[B(AB)^{-1}] = I$. Thus A is invertible, with inverse $B(AB)^{-1}$. Similarly, $(AB)^{-1}(AB) = I$ gives $[(AB)^{-1}A]B = I$. Thus B is invertible, with inverse $(AB)^{-1}A$.

6.3.2 An Algorithm for Finding the Inverse

PROPOSITION 6.4

A square matrix A is invertible if and only if it is row-equivalent to the identity matrix.

Algorithm

1. Form $(A \mid I)$ and row-reduce, aiming to put the left block into **reduced row-echelon form**.
2. Inspect the resulting left block:
   - If it is I, then A is invertible, and the right block is its inverse: $(I \mid A^{-1})$.
   - If it is not I, then A is not invertible.

EXERCISE 6.7

Please determine whether the following matrices are invertible. For each invertible matrix, please find its inverse by row reduction and please verify the result:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & -4 \\ 3 & -6 \end{pmatrix}.$$

WORKED SOLUTION

Row-reduce $(A \mid I_3)$:

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

After $R_3 \leftarrow R_3 - R_2$, this becomes

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

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EXERCISE 6.7 CONTINUED

Finally, $R_1 \leftarrow R_1 - 3R_3$ gives

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & -3 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

Hence $A^{-1} = \begin{pmatrix} 3 & 1 & -3 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$. To verify, left multiplication by A forms, respectively, the second row of the inverse, its first row plus three times its third row, and the sum of its second and third rows:

$$AA^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 3-3 & 1 & -3+3 \\ 1-1 & 0 & 1 \end{pmatrix} = I_3.$$

For B, the second row is $3/2$ times the first. Thus $R_2 \leftarrow R_2 - (3/2)R_1$ creates a zero row, so B is not invertible.

6.4 Elementary Matrices

DEFINITION 6.4

- Elementary matrices are square matrices representing the elementary row operations used in row reduction.
- Applying an elementary row operation to a matrix is equivalent to multiplying it **on the left** by the corresponding elementary matrix.
- Each elementary matrix is obtained by applying its row operation to the identity matrix.

Consequence. Matrices A, B are row-equivalent if and only if there is a finite sequence of elementary matrices such that $B = E_1 E_2 \cdots E_k A$.

1. **Interchanging rows** i and j is represented by $E^{(1)}_{i,j}$:

$$E^{(1)}_{i,j} = \begin{pmatrix} 1 & & & & & \\ & 0 & & 1 & & \\ & & 1 & & & \\ & 1 & & 0 & & \\ & & & & 1 & \\ & & & & & \ddots \end{pmatrix}.$$

   - This matrix is its own inverse: interchange the same two rows again.

2. **Multiplying row** i by a nonzero scalar λ is represented by $E^{(2)}_{i,\lambda}$:

$$E^{(2)}_{i,\lambda} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

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- Its inverse is $E^{(2)}_{i,\lambda^{-1}}$: multiply the same row by the reciprocal scalar.

3. **Adding λ times row j** to row i, with $i \neq j$, is represented by $E^{(3)}_{i,j,\lambda}$. The entry λ is in row i, column j:

$$E^{(3)}_{i,j,\lambda} = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & \lambda & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix}.$$

- Its inverse is $E^{(3)}_{i,j,-\lambda}$: add $-\lambda$ times row j to row i.

EXERCISE 6.8

Please write $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix}$ as a product of elementary matrices. Please use the row reduction from the previous exercise and please check the order of the factors.

WORKED SOLUTION

The three row operations interchange R_1, R_2, subtract R_2 from R_3, and subtract $3R_3$ from R_1. Their matrices are

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplication is on the left, so $E_3E_2E_1A = I_3$. Multiplying by the inverses in the reverse order yields

$$A = E_1^{-1}E_2^{-1}E_3^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Indeed, the last two factors multiply to $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, and the first interchanges the first two rows to recover A. Each of the three factors represents one elementary row operation.

EXERCISE 6.9

Let $A, B, C \in \mathbb{R}^{n\times n}$ be invertible, and let $\alpha \in \mathbb{R}$ with $\alpha \neq 0$. Define

$$D := \alpha A^2 B^T C^3 A.$$

1. Please prove that D is invertible.
2. Please find an explicit expression for D^{-1} in terms of A, B, C, α.

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EXERCISE 6.9 CONTINUED

WORKED SOLUTION

1. Each of $\alpha I_n, A, A^2, B^T, C^3$ is invertible:
   - $\alpha \neq 0$, so αI_n is invertible.
   - A is invertible by assumption, so A^2 is invertible.
   - Since B is invertible, B^T is invertible with $(B^T)^{-1} = (B^{-1})^T$.
   - Since C is invertible, C^3 is invertible with $(C^3)^{-1} = (C^{-1})^3$.

   A product of invertible matrices is invertible, so D is invertible.

2. Write the ordered product $D = \alpha A^2 B^T C^3 A$. The inverse of a product is the product of the inverses in reverse order:

$$(XYZ)^{-1} = Z^{-1}Y^{-1}X^{-1}.$$

   Thus

$$D^{-1} = A^{-1}(C^3)^{-1}(B^T)^{-1}(A^2)^{-1}(\alpha I_n)^{-1}.$$

   Compute each factor:

$$(C^3)^{-1} = (C^{-1})^3, \quad (B^T)^{-1} = (B^{-1})^T, \quad (A^2)^{-1} = (A^{-1})^2, \quad (\alpha I_n)^{-1} = \frac{1}{\alpha} I_n.$$

   Substitution gives

$$D^{-1} = A^{-1}(C^{-1})^3(B^{-1})^T(A^{-1})^2 \cdot \frac{1}{\alpha} I_n = \frac{1}{\alpha} A^{-1}(C^{-1})^3(B^{-1})^T(A^{-1})^2.$$

   This is the required expression for D^{-1}.

EXERCISE 6.10

Let

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Please prove that A is invertible if and only if every $\lambda_i \neq 0$, and please find an explicit formula for A^{-1}.

WORKED SOLUTION

If A is invertible, then every $\lambda_i \neq 0$. Suppose A is invertible and, for a contradiction, that

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EXERCISE 6.10 CONTINUED

$\lambda_k = 0$ for some index k. Consider the standard vector

$$\vec{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

with 1 in position k and zero in every other coordinate. Then

$$A\vec{e}_k = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \lambda_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \vec{0},$$

since $\lambda_k = 0$. Thus $A\vec{e}_k = \vec{0}$ with $\vec{e}_k \neq \vec{0}$, a nontrivial solution of $Ax = 0$. If A were invertible, left multiplication by A^{-1} would give $e_k = A^{-1}0 = 0$, a contradiction. Therefore every $\lambda_i \neq 0$. **If every $\lambda_i \neq 0$, then A is invertible; we find its inverse.** Assume all these scalars are nonzero. Define

$$B := \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda_n} \end{pmatrix}.$$

Calculate

$$\begin{aligned} AB &= \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}\left(\lambda_1 \frac{1}{\lambda_1}, \ldots, \lambda_n \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}(1, \ldots, 1) = I_n. \end{aligned}$$

The same calculation in reverse order gives $BA = I_n$. Hence B is the inverse of A, so

$$A^{-1} = B = \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right).$$

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6.4.1 Equivalent Conditions

PROPOSITION 6.5

For a square matrix $A \in M_n(\mathbb{F})$, the following conditions are equivalent:

1. A is invertible.
2. A is row-equivalent to I_n.
3. $\text{rank}(A) = n$.
4. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a solution.
5. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a unique solution $\vec{x} \in \mathbb{F}^n$.

6.4.2 The Connection with the Homogeneous System

In the preceding assertions, take $\vec{b} = \vec{0}$. There is then a unique solution, the trivial solution. Thus:

PROPOSITION 6.6

Another equivalent condition: $A \in M_n(\mathbb{R})$ is invertible if and only if its homogeneous system has only the trivial solution. In other words, the only solution of $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$.

EXERCISE 6.11

For $\alpha, \gamma, \phi \in \mathbb{R}$, consider $Ax = b$, where

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = (\alpha, 4\gamma, 5\phi)^T.$$

1. Please express the solution in terms of the parameters.
2. Please prove that the system is homogeneous if and only if its unique solution is zero.

WORKED SOLUTION

Using the inverse calculated earlier,

$$x = A^{-1}b = \begin{pmatrix} 3 & 1 & -3 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ 4\gamma \\ 5\phi \end{pmatrix} = \begin{pmatrix} 3\alpha + 4\gamma - 15\phi \\ \alpha \\ 5\phi - \alpha \end{pmatrix}.$$

The coordinates are obtained from the sums $3\alpha + 1 \cdot 4\gamma - 3 \cdot 5\phi$, α, and $-\alpha + 5\phi$. Direct substitution in the equations gives $x_2 = \alpha$, $x_1 + 3x_3 = 4\gamma$, and $x_2 + x_3 = 5\phi$. Because A is invertible, this solution is unique. The system is homogeneous exactly when $b = 0$, that is, when $\alpha = \gamma = \phi = 0$; the formula then gives $x = 0$. Conversely, if $x = 0$ is a solution, then $b = A0 = 0$, so the system is homogeneous. Invertibility ensures uniqueness in this case as well.

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EXERCISE 6.12

Three channels are coupled by $A = I_3 + N$, where

$$N = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Please compute N^2, N^3, please find A^{-1} without row reduction, and reconstruct the state from $y = Ax = (7, 4, 2)^T$.

WORKED SOLUTION

The only nonzero entry of N^2 is $(N^2)_{13} = N_{12}N_{23} = 2 \cdot 1 = 2$. Hence

$$N^2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

Compute the products in both orders:

$$(I + N)(I - N + N^2) = I - N + N^2 + N - N^2 + N^3 = I,$$

$$(I - N + N^2)(I + N) = I + N - N - N^2 + N^2 + N^3 = I.$$

Therefore

$$A^{-1} = I - N + N^2 = \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Now

$$x = \begin{pmatrix} 7 - 2 \cdot 4 + 2 \cdot 2 \\ 4 - 2 \\ 2 \end{pmatrix} = (3, 2, 2)^T.$$

Verification: $Ax = (3 + 2 \cdot 2, 2 + 2, 2)^T = (7, 4, 2)^T$. Three applications of the coupling give zero, so the inverse series terminates.

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Additional practice

EXERCISE 6.13 FOUNDATION

Over $\mathbb{R}$, let $w_1 = (1, 1, 0)^T$ and $w_2 = (0, 1, 1)^T$.

1. Please show that adding $e_3 = (0, 0, 1)^T$ extends the pair to an ordered basis $B = (w_1, w_2, e_3)$ of $\mathbb{R}^3$.
2. Please use row reduction to find the inverse of the matrix A whose columns are the members of B.
3. Please compute $[b]_B$ for $b = (2, 5, 7)^T$ and please verify the representation.

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EXERCISE 6.13 FOUNDATION CONTINUED

WORKED SOLUTION

If $aw_1 + bw_2 + ce_3 = 0$, comparison of coordinates gives $a = 0$, $a + b = 0$, and $b + c = 0$, so $a = b = c = 0$. Three independent vectors in $\mathbb{R}^3$ form a basis; in particular, the first two are independent as well.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Row-reduce the augmented matrix:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right).$$

Thus

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}, \qquad [b]_B = A^{-1}b = \begin{pmatrix} 2 \\ -2+5 \\ 2-5+7 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}.$$

Indeed, $2w_1 + 3w_2 + 4e_3 = (2, 2+3, 3+4)^T = (2, 5, 7)^T$. Both products $AA^{-1}, A^{-1}A$ equal I_3, and the representation is unique because B is a basis.

EXERCISE 6.14 CHALLENGE

Let $u, v \in \mathbb{R}^n$, $n \geq 1$, and define $A = I_n + uv^T$.

1. Please prove that A is invertible if and only if $1 + v^T u \neq 0$, and please find its inverse in this case without using a determinant.
2. Please apply the formula to $u = (1, 1, 0)^T$, $v = (1, 0, 1)^T$, and please solve $Ax = (5, 7, 1)^T$.

WORKED SOLUTION

Write $P = uv^T$ and $c = v^T u \in \mathbb{R}$. Associativity gives

$$P^2 = (uv^T)(uv^T) = u(v^T u)v^T = cP.$$

If $1 + c \neq 0$, define $B = I_n - P/(1+c)$. Then

$$AB = (I_n + P)\left(I_n - \frac{P}{1+c}\right) = I_n + \left(1 - \frac{1}{1+c} - \frac{c}{1+c}\right)P = I_n.$$

Multiplication in the opposite order gives exactly the same expansion, so $BA = I_n$. Therefore

$$A^{-1} = I_n - \frac{uv^T}{1 + v^T u}.$$

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EXERCISE 6.14 CHALLENGE CONTINUED

If $1 + c = 0$, then $c = -1$, so $u \neq 0$, whereas

$$Au = u + u(v^T u) = (1 + c)u = 0.$$

An inverse would imply $u = 0$, a contradiction. This proves both directions.
In the example, $v^T u = 1$, so

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}.$$

Hence

$$x = A^{-1} \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix} = \begin{pmatrix} (5-1)/2 \\ -5/2 + 7 - 1/2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}.$$

Direct verification: $Ax = (2 \cdot 2 + 1, 2 + 4 + 1, 1)^T = (5, 7, 1)^T$.

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EXERCISE 6.15 CHALLENGE

Let $A \in M_n(\mathbb{R})$ satisfy $A^m = I_n$ for an integer $m \geq 2$, and define

$$P = \frac{1}{m} \sum_{k=0}^{m-1} A^k.$$

Please prove that $AP = PA = P$ and $P^2 = P$. Please find the inverse of $I_n + P$.

WORKED SOLUTION

Multiplication by A shifts the powers, so

$$AP = \frac{1}{m}(A + A^2 + \cdots + A^m) = P.$$

Since powers of A commute with A, we also have $PA = P$. Induction gives $A^k P = P$ for every $k \geq 0$. Thus

$$P^2 = \frac{1}{m} \sum_{k=0}^{m-1} A^k P = P.$$

Consequently

$$(I_n + P)(I_n - P/2) = I_n + P/2 - P^2/2 = I_n.$$

The reverse product is also I_n, since both factors are polynomials in P. Therefore $(I_n+P)^{-1} = I_n - P/2$.

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EXERCISE 6.16 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ and define the block matrix

$$C = \begin{pmatrix} A & B \\ B & A \end{pmatrix}.$$

Please prove that C is invertible if and only if $A+B$ and $A-B$ are invertible. In that case, please find C^{-1} in terms of $(A+B)^{-1}$ and $(A-B)^{-1}$.

WORKED SOLUTION

Define

$$J = \begin{pmatrix} I_n & I_n \\ I_n & -I_n \end{pmatrix}.$$

Since $J^2 = 2I_{2n}$, we have $J^{-1} = J/2$. Block multiplication gives

$$CJ = \begin{pmatrix} A+B & A-B \\ A+B & -(A-B) \end{pmatrix},$$

and therefore

$$J^{-1}CJ = \begin{pmatrix} A+B & 0 \\ 0 & A-B \end{pmatrix}.$$

Multiplication on either side by invertible matrices preserves invertibility. A block diagonal matrix is invertible exactly when both its diagonal blocks are invertible. This proves the equivalence. Writing $X = (A+B)^{-1}$ and $Y = (A-B)^{-1}$, we obtain

$$\begin{aligned} C^{-1} &= J \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix} J^{-1} \\ &= \frac{1}{2} \begin{pmatrix} X+Y & X-Y \\ X-Y & X+Y \end{pmatrix}. \end{aligned}$$

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